

Alireza Safarzadeh

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PROFESSIONAL SUMMARY

Data Scientist / BI Developer with 5+ years of experience in fintech analytics, transaction monitoring, predictive modeling, business intelligence, and data warehousing. Skilled in Python (pandas, NumPy, scikit-learn), TensorFlow/Keras, SQL, SSIS, Power BI, and end-to-end data pipelines. Built production models for ATM status prediction, customer churn, and medical-image segmentation, and developed dimensional data warehouses and KPI dashboards for holding-company performance monitoring. Experienced in reliability analytics on banking switch and core systems, ETL automation, and KPI reporting (availability, MTTR, MTTF).

CORE SKILLS

- **Programming:** Python, SQL, MATLAB
- **BI & Data Warehousing:** Dimensional modeling, fact/dimension design, Power BI dashboards, KPI reporting
- **Data Engineering:** ETL pipelines, SSIS, web crawling/scraping (Selenium, BeautifulSoup/bs4), log processing, Oracle/SQL Server views, batch automation
- **ML:** Ensemble learning (RF, GBM, LightGBM, CatBoost), SHAP, cross-validation, feature engineering
- **Deep Learning:** TensorFlow/Keras, UNet/UNet++, CNN segmentation, mixed precision, EMA, TTA
- **Visualization:** Power BI, Matplotlib, Seaborn, Plotly
- **Domains:** Fintech (transaction routing, switch/core analytics, ATM reliability), holding/subsidiary performance monitoring, medical imaging

INDUSTRY EXPERIENCE

Arzesh Afarinan Fadak Holding Co.

BI Developer

Tehran, Iran

Dec 2025 – Present

Projects

Holding Performance Data Warehouse & BI Dashboards

Dec 2025 – Present

- Designed and implemented a dimensional data warehouse with fact and dimension tables to monitor the performance of subsidiary companies across the holding.
- Built web-crawling pipelines with Selenium and BeautifulSoup (bs4) to collect operational data from Railways of the Islamic Republic of Iran websites.
- Developed SSIS ETL packages to extract, clean, transform, and load data from multiple sources into the warehouse.
- Created Power BI dashboards and management reports to monitor KPIs, trends, and subsidiary-company performance.

Pulseware Co.

Senior Data Analyst

Junior Data Analyst

Data Analyst Intern

Tehran, Iran

Mar 2023 – Dec 2025

Feb 2020 – Mar 2023

Nov 2019 – Feb 2020

Projects

Switch & Core Banking Reliability Analytics

2019 – Dec 2025

- Analyzed operational logs from the bank's internal switch and core banking system to measure uptime, downtime, outage frequency, and failure patterns over monthly and yearly periods.
- Computed reliability KPIs—including availability, MTTR, MTTF, and failure counts—for both switch and core components.
- Correlated switch-level failures with core-banking responses to identify whether outages originated from the switch or the core platform.
- Developed automated scripts to detect downtime intervals, quantify transaction impact, and categorize failures by component.
- Delivered SLA-focused reliability reports for governance teams and IT operations.

e-Government API Performance & Reliability Analytics

2020 – Dec 2025

- Analyzed high-volume API call logs for national verification services including Postal Address Verification, National Civil Registry (ID validation), and Shahkar mobile-ID matching.
- Measured availability, timeout frequency, error rates, and latency across banking channels (branches, ATM, mobile banking).
- Generated monthly SLA/compliance reports to evaluate provider performance and detect degradation patterns.

ATM Status Error Reduction with ML (with dashboard integration)

Nov 2019 – Dec 2025

- Parsed ATM journal text files at scale; engineered features (e.g., exponential interval features) to improve status detection.
- Evaluated multiple models (Decision Tree, Bagging, LightGBM, CatBoost, DES/DCS, Random Forest); selected robust ensembles for deployment.
- Integrated model outputs into the bank's operational dashboard to reduce false alarms and missed outages.

Automated ATM Network Quality Reporting

Sep 2021 – Dec 2025

- Designed ETL pipelines to extract data from SQL/Oracle views aligned to reporting needs.
- Automated monthly KPI computation (availability, MTTR, MTTF) using MATLAB and integrated the workflow into internal reporting.

Peak Transaction Forecasting (Nowruz Period)

Jan 2024 – Mar 2024

- Analyzed historical March transaction patterns; engineered temporal/seasonal features and evaluated multiple models.
- Selected Cubic SVM for peak forecasting based on validation performance.

Interactive RTGS Transaction Analyzer (MATLAB UI)

Jan 2022 – Mar 2022

- Developed an interactive MATLAB UI to select banks and visualize RTGS volumes using dynamic race charts.

EDUCATION

University of Tehran

Sep 2018 – Feb 2022

M.Sc. in Electrical Engineering (Control)

Tehran, Iran

- Thesis: *Optimizing False Alarms in Automated Teller Machines (ATMs) with Data Fusion Methods*
- Industry Partner: Pulseware Co.

Zanjan University

Sep 2013 – Sep 2018

B.Sc. in Electrical Engineering (Control)

Zanjan, Iran

PROJECTS

Brain Tumor Segmentation with Multi-Model Fusion

2024 – Present

- Implemented a semantic segmentation pipeline on the Brain Tumor MRI dataset using UNet, UNet++, DeepLab, and MobileNet.
- Developed a UNet-based fusion refiner combining multi-model probability maps, achieving Dice = 0.852 and IoU = 0.743.
- Applied Albumentations, mixed precision training, EMA, TTA, BCE+Dice loss, and post-processing (connected component cleanup).

ISEPS: Customer Churn Prediction for Financial Institutions

2024 – 2025

- Built a stacked ensemble (RF, GBM, LightGBM, CatBoost) on a real 10,000-customer dataset.
- Engineered RFM, Balance per Product, Engagement Score, anomaly scores; handled imbalance with ADASYN.
- Improved F1-score from 80% (best single model) to 95% with SHAP-based interpretability.

Optimizing False Alarms in ATMs (M.Sc. Thesis)

2019 – 2022

- Industrial project improving ATM status detection using ensembles and log-derived features.

PUBLICATIONS

A. Azarsina, A. Safarzadeh, M. R. Jamali, and A. H. Vahabie, “Adaptive Pattern Reconstruction Using Linear Regression for Improved TPS Anomaly Detection,” in *2025 15th International Conference on Computer and Knowledge Engineering (ICCKE)*, 2025, pp. 1–5. [DOI link](#)

A. Safarzadeh, M. R. Jamali, and B. Moshiri, “Enhancing Precision of Automated Teller Machines Network Quality Assessment: Machine Learning and Multi Classifier Fusion Approaches,” arXiv preprint arXiv:2501.01067, 2025. [arXiv link](#)

A. Safarzadeh, M. J. Heidarian, R. Sharifanfar, S. F. Haqiqi, and M. R. Jamali, “Estimating the Trend of COVID-19 in Iran Before and After the Start of Vaccination,” in *2021 International Conference on Computational Science and Computational Intelligence (CSCI)*, 2021, pp. 1209–1216. [DOI link](#)

TEACHING

Teaching Assistant, University of Tehran

- Linear Control Systems Lab

Sep 2019 – Feb 2020

INTERESTS

Film, Badminton, Setar, Running.